

Every rule the backend actually enforces

Three domains, 231 rules — what you can't do, what triggers a warning instead of a block, and why. Reference this before assuming a behavior is a bug: most of what looks like a gap turns out to already be a documented, deliberate rule below.

✖ **HARD BLOCK** — the action is rejected outright, nothing changes.

⚠ **SOFT SIGNAL** — the action proceeds; a flag, notice, or downstream effect follows.

Vehicle CRUD, readiness scoring, categories, issue reports, maintenance records, schedules, condition checks, documents, hubs, and locations.

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- 01 Only Admin can create, update, archive, restore, or decommission a vehicle.
× HARD BLOCK FleetController.php:788,815,836,861,901
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- 02 A vehicle cannot be archived while it has open (non-Closed/Cancelled) tickets.
Why — close or cancel the open ticket(s) first — the response lists exactly which ones.
× HARD BLOCK FleetController.php:838-846
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- 03 A vehicle already Decommissioned cannot be decommissioned again.
× HARD BLOCK FleetController.php:903
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- 04 A vehicle cannot be decommissioned while it has open tickets — same rule as archiving.
Why — a beyond-repair vehicle's unfinishable ticket should go through decision-close first.
× HARD BLOCK FleetController.php:905-913
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- 05 Decommissioning requires a mandatory reason.
× HARD BLOCK FleetController.php:915-917
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- 06 Only an Inactive or Decommissioned vehicle can be restored.
Why — Restore clears both archive and decommission fields at once, since it reverses either state.
× HARD BLOCK FleetController.php:866-870
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- 07 A readiness check cannot be recorded for an Inactive/Decommissioned vehicle.
× HARD BLOCK FleetController.php:943-947
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- 08 The "mark Available" shortcut after a readiness check only appears when every checklist item passed, the vehicle isn't already Available, and it has no open ticket.
Why — a vehicle with an open ticket must return to Available through that ticket closing, not a generic checklist.
△ SOFT SIGNAL FleetController.php:971-979
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- 09 markVehicleAvailable cannot be used on an Inactive/Decommissioned vehicle.
× HARD BLOCK FleetController.php:1000-1004
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- 10 markVehicleAvailable cannot be used while the vehicle has an open ticket — re-checked independently in case state changed since the checklist was shown.
× HARD BLOCK FleetController.php:1006-1009
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- 11 Plate numbers must be unique fleet-wide and match a PH-style plate pattern.
- × **HARD BLOCK** FleetController.php:2526-2539
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- 12 Hull material and engine type are required only for Water-domain categories, optional for Land.
- × **HARD BLOCK** FleetController.php:2547-2548
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- 13 Current location must be one of the registered hub names.
- × **HARD BLOCK** FleetController.php:1060,2550
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- 14 New vehicles always start Available / Good — not settable at creation.
- △ **SOFT SIGNAL** FleetController.php:799-802
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- 15 "In Use" is intentionally absent from the status list — this system tracks availability only, nothing ever sets a vehicle to In Use.
- △ **SOFT SIGNAL** FleetController.php:71-73

01 Readiness state is derived, never stored, in strict order: retired → in maintenance → not-ready (bad condition) → never checked → not-ready (failed check) → stale (check too old) → ready.

Why — "Available" status alone is necessary but not sufficient for "ready to respond."

△ **SOFT SIGNAL**

FleetController.php:430-451

02 A passing readiness check stays fresh for 24 hours only — after that it reads "stale, needs re-check."

△ **SOFT SIGNAL**

FleetController.php:38-39,448-450

03 A vehicle category with zero ready units among its active vehicles is flagged "no coverage" on the dashboard.

△ **SOFT SIGNAL**

FleetController.php:509,483

04 A category with 1 or fewer ready units is flagged "single point of failure"; 0 ready is additionally "critical."

Why — keyed off ready count, not fleet size — a 3-unit category with 2 in the shop still triggers it.

△ **SOFT SIGNAL**

FleetController.php:563-566

05 "Chronic" vehicle = 3+ tickets in 6 months, or any ticket flagged a recurrence of a prior fix.

△ **SOFT SIGNAL**

FleetController.php:650-653

06 "Decommission signal" = lifetime repair spend at or above 60% of the vehicle's acquisition cost (only computed when cost is on file).

△ **SOFT SIGNAL**

FleetController.php:659-662

07 Preventive maintenance is "due soon" inside a 7-day window; anything scheduled before today is "overdue."

△ **SOFT SIGNAL**

FleetController.php:596,612

08 The dashboard Action Queue has a fixed urgency order: critical readiness first, then close-ready tickets, unassigned sub-issues, pending confirmations, recurring-fault reviews, pending issues, readiness checks, then overdue schedules.

△ **SOFT SIGNAL**

FleetController.php:393-411

09 A ticket is flagged for "recurring fault review" once it's the 3rd occurrence of the same fault within the recurrence window, and stays flagged for as long as it's open.

△ **SOFT SIGNAL**

FleetController.php:383-391

01 Only Admin can create, update, or delete a vehicle category.

× **HARD BLOCK**

FleetController.php:686,702,723

02 Category names must be unique.

× **HARD BLOCK**

FleetController.php:689,705-709

03 A category cannot be deleted while any vehicle is still assigned to it.

× **HARD BLOCK**

FleetController.php:725

04 A category's domain (Land/Water) must be one of those two values, and it drives the hull material/engine type requirement on vehicles.

× **HARD BLOCK**

FleetController.php:690,711

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- 01 Cannot report a new issue on an archived or decommissioned vehicle.
✖ HARD BLOCK FleetController.php:1276-1282
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- 02 Filing an issue immediately sets the vehicle's condition to "Needs Inspection."
⚠ SOFT SIGNAL FleetController.php:1297-1299
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- 03 Maintenance Personnel may file issues, but only as part of self-documenting a field repair, not as a general reporting permission.
⚠ SOFT SIGNAL FleetController.php:1258-1262
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- 04 A Custodian may only edit or delete their own issue reports — Admin and Maintenance Personnel aren't restricted this way.
✖ HARD BLOCK FleetController.php:1314-1315,1387-1388
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- 05 A Custodian can only edit/delete their issue while it's still Pending — once reviewed, it locks from the reporter.
✖ HARD BLOCK FleetController.php:1316,1389
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- 06 Changing an issue's status drives automatic vehicle status/condition changes: In Maintenance → Under Maintenance/Needs Repair; Resolved → Available/Good; Pending/Under Review → Available/Needs Inspection.
⚠ SOFT SIGNAL FleetController.php:1360-1369
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- 07 Issues on retired vehicles are excluded from the issue list entirely.
⚠ SOFT SIGNAL FleetController.php:1227-1231
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- 08 Duplicate-issue detection is a non-blocking warning only — the endpoint just returns open reports on a vehicle so the UI can flag it before submission.
⚠ SOFT SIGNAL FleetController.php:1400-1404
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- 09 Viewing a single issue is deliberately open to any Admin/Custodian/Maintenance Personnel, not scoped to "my issues" — the duplicate-warning link has to show someone else's report.
⚠ SOFT SIGNAL FleetController.php:1190-1201

01 A record must name who performed the repair — a registered user or free-text "performed by other," never blank.

× **HARD BLOCK**

FleetController.php:1584-1588

02 A non-Admin submitter is always recorded as having performed the repair themselves; only Admin can attribute it to someone else.

△ **SOFT SIGNAL**

FleetController.php:1579-1582

03 A vehicle can't be listed as the source of its own cannibalized part.

× **HARD BLOCK**

FleetController.php:1544,1565,1698,1716

04 Creating a record already marked Completed is Admin-only.

× **HARD BLOCK**

FleetController.php:1601

05 Fast-closing to Completed additionally requires proof attached — a receipt or photo, since a typed note "proves nothing a Custodian couldn't just as easily fabricate."

Why — this is the only path to a Completed record at creation — everything else runs the full Assign → Verify → Confirm chain.

× **HARD BLOCK**

FleetController.php:1602-1607

06 A non-Admin who tries to mark their own record Completed on update is silently downgraded to "For Verification" instead — the request succeeds, the outcome is coerced.

△ **SOFT SIGNAL**

FleetController.php:1734-1741

07 Marking a record Completed via update is Admin-only, and needs either the receipt fast-close proof or a Passed verification result.

× **HARD BLOCK**

FleetController.php:1752-1758

08 A record can only be verified while status is "For Verification."

× **HARD BLOCK**

FleetController.php:1834-1838

09 A record can only be verified once — a second verdict is blocked until an Admin explicitly reopens it.

Why — prevents a duplicate submission silently overwriting the first verdict.

× **HARD BLOCK**

FleetController.php:1846-1850

- 10 A Custodian cannot verify a repair they personally performed — someone else must sign off, even on a dual-role account.
Why — the core "don't grade your own homework" rule for maintenance records.
× **HARD BLOCK** FleetController.php:1856-1860
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- 11 Admin's final confirmation only works on a record that's "For Verification" and already Passed Custodian verification.
× **HARD BLOCK** FleetController.php:1901-1905
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- 12 Rejecting confirmation reopens the record — status back to Under Repair, verification verdict cleared — so it can pass through the once-only verify rule again later.
△ **SOFT SIGNAL** FleetController.php:1938-1949
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- 13 Decision-close (skip verification) cannot be used on an already-Completed record.
× **HARD BLOCK** FleetController.php:1986-1990
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- 14 Decision-close cannot be used once verification has already Passed — that must go through Confirm instead, never get overwritten by an unverified close.
× **HARD BLOCK** FleetController.php:1995-1999
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- 15 Decision-close requires a mandatory closure reason and an explicit fit-for-service yes/no — never assumed.
Why — verification_result is deliberately left null so a decision-close can never masquerade as a passed verification.
× **HARD BLOCK** FleetController.php:2001-2007
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- 16 Decision-close only resolves the originating issue report if the vehicle was marked fit for service — otherwise the issue stays open since the underlying problem still stands.
△ **SOFT SIGNAL** FleetController.php:2028-2033
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- 17 Cannibalizing a part is logged on the donor vehicle but never auto-changes its status/condition — deliberately left as "a human judgment call."
△ **SOFT SIGNAL** FleetController.php:1653-1658
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- 18 A non-Admin filing a standalone record with no prior ticket triggers an Admin notification, since Admin wasn't in the loop when the work started.
△ **SOFT SIGNAL** FleetController.php:1674-1685

- 01 Only one active schedule is allowed per vehicle per date — enforced on create, update, and restore alike.

Why — a second one on the same date is almost always a mistake.

✖ **HARD BLOCK** FleetController.php:2115-2119, 2177-2182, 2405-2410

- 02 New schedules always start "Scheduled" — Completed/Cancelled are only reachable through their own actions, never set directly at creation.

⚠ **SOFT SIGNAL** FleetController.php:2123-2126

- 03 Update can't set status straight to Completed — only through the dedicated complete action, so the linked record and recurrence still get created.

✖ **HARD BLOCK** FleetController.php:2161-2166

- 04 Completing a schedule only works while it's still "Scheduled."

✖ **HARD BLOCK** FleetController.php:2216

- 05 A non-Admin can only complete a schedule assigned to themselves — an unassigned schedule can't be completed by a non-Admin at all.

Why — otherwise two mechanics could show up for the same job.

✖ **HARD BLOCK** FleetController.php:2224-2232

- 06 Completing a schedule routes the resulting record to "For Verification," never straight to Completed — closes the one path that used to let routine PM self-certify.

⚠ **SOFT SIGNAL** FleetController.php:2265-2273

- 07 Fast-closing a schedule completion is Admin-only and still requires proof attached — a mechanic can attach proof but can't self-certify past Custodian verification.

✖ **HARD BLOCK** FleetController.php:2253-2260

- 08 A recurring schedule auto-generates its next occurrence at completion, nudging the date forward (up to 60 days) if the natural next date collides with the one-per-day rule.

⚠ **SOFT SIGNAL** FleetController.php:2326-2346

- 09 Deleting a schedule is always a soft-cancel — the row never actually disappears.

⚠ **SOFT SIGNAL** FleetController.php:2375-2384

- 10 Restoring only works on a Cancelled schedule, and is subject to the same one-per-vehicle-per-date rule.

✖ **HARD BLOCK** FleetController.php:2396-2410

- 11 Create is open to Admin, Maintenance Personnel, and Custodian ("proposing when" isn't a commitment); update/complete/delete/restore are Admin and Maintenance Personnel only.

△ **SOFT SIGNAL**

FleetController.php:2096,2151,2214,2377,2394

CONDITION CHECKS — 3

- 01 A "Needs Repair" or "Damaged" result automatically flips the vehicle to Under Maintenance — applied consistently on create, update, and delete.

△ **SOFT SIGNAL**

FleetController.php:1104-1106,1138-1140,1168-1171

- 02 Deleting a condition check re-derives the vehicle's condition from whatever check is now the latest, or "Good" if none remain.

△ **SOFT SIGNAL**

FleetController.php:1163-1181

- 03 Only Admin/Custodian may create, update, or delete condition checks.

× **HARD BLOCK**

FleetController.php:1091,1124,1157

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- 01 Admin, Custodian, and Maintenance Personnel can all add/update/delete vehicle documents — no ownership restriction, unlike issue reports.
- △ **SOFT SIGNAL** FleetController.php:1431,1459,1480
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- 02 Documents are kept deliberately separate from the vehicle's cover photo — receipts/registration/insurance are a file cabinet, not the profile image.
- △ **SOFT SIGNAL** FleetController.php:1417-1420
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- 03 Only Admin can create, update, or delete a hub.
- × **HARD BLOCK** HubController.php:27,55,74
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- 04 Hub names must be unique.
- × **HARD BLOCK** HubController.php:30,58
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- 05 Default hubs can't be deleted, only hidden.
- × **HARD BLOCK** HubController.php:76
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- 06 Only Admin can log a location update, which both records the location and overwrites the vehicle's current location.
- × **HARD BLOCK** FleetController.php:1056,1065-1068

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- 01 Every "notify role X" helper is scoped to the same barangay — nobody gets notified across tenants.
-  **SOFT SIGNAL** FleetController.php:2593-2596,2599,2613
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- 02 Staff lookups are manually filtered by the requester's barangay, since the User model has no automatic scope — otherwise a Custodian could assign a mechanic from a different barangay.
-  **HARD BLOCK** FleetController.php:44-48,56-59
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- 03 "Dispatchable mechanics" (Maintenance Personnel only) and "anyone who could have performed a repair" (Admin/Custodian too) are deliberately different lists.
-  **SOFT SIGNAL** FleetController.php:56-67
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- 04 A Custodian's dashboard badges count only their own reported issues, not the whole fleet.
-  **SOFT SIGNAL** FleetController.php:164-173,119-125
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- 05 A Maintenance Personnel's schedule badge counts only their own assigned upcoming schedules.
-  **SOFT SIGNAL** FleetController.php:177-185
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- 06 Vehicle status/condition is re-derived from its latest issue and any active ticket/record on every dashboard load — not a background job.
-  **SOFT SIGNAL** FleetController.php:2636-2703

The 5-phase maintenance ticket lifecycle: creation, inspection, work orders, verification, and closing — plus cancel, delete, defer, and reopen.

01 Only Admin can create a ticket.

× **HARD BLOCK**

TicketController.php:160

02 Cannot open a ticket on an archived or decommissioned vehicle.

Why — a retired vehicle is out of the fleet — no new work goes on it.

× **HARD BLOCK**

TicketController.php:199-205

03 Cannot open a duplicate ticket for the same main issue on the same vehicle while an open one already exists — titles are compared after stripping "[Issue #N]" prefixes so rewording can't dodge it.

Why — suggests adding a sub-issue to the existing ticket instead.

× **HARD BLOCK**

TicketController.php:207-222

04 The assigned custodian must actually hold the Custodian role, and belong to the same barangay as the vehicle.

× **HARD BLOCK**

TicketController.php:225-226

05 Sub-issues are required when entry mode is "pre-diagnosed."

× **HARD BLOCK**

TicketController.php:176

06 Fault category defaults from the linked Issue Report when the Admin didn't supply one, so recurrence/cost grouping always has something to key on.

△ **SOFT SIGNAL**

TicketController.php:228-234

07 Recurrence detection counts the same fault fixed on this vehicle in the last 90 days (by category or normalized title) — flags "Recurring Fault Detected" but never blocks creation.

△ **SOFT SIGNAL**

TicketController.php:236-241,322-340

08 A pre-diagnosed ticket is born Active with inspection auto-stamped "Needs Maintenance" — inspection is skipped, not faked.

△ **SOFT SIGNAL**

TicketController.php:255-293

09 An inspection-mode ticket starts Open, awaiting the Custodian.

△ **SOFT SIGNAL**

TicketController.php:257

10 Creating a ticket from a linked Issue Report moves that report to "In Maintenance."

△ **SOFT SIGNAL**

TicketController.php:295-299

01 Only Custodian role may submit an inspection.

× **HARD BLOCK**

TicketController.php:354

02 Only the specific Custodian assigned to this ticket may submit its inspection — not any Custodian.

× **HARD BLOCK**

TicketController.php:356

03 Inspection can only be submitted while the ticket is Open.

× **HARD BLOCK**

TicketController.php:357

04 Sub-issues are required when the inspection result is "Needs Maintenance."

× **HARD BLOCK**

TicketController.php:362

05 Confirming a real problem immediately flips the vehicle to Needs Repair / Under Maintenance, whether or not a mechanic is assigned yet.

△ **SOFT SIGNAL**

TicketController.php:395-402

06 A "No Issues" result clears the vehicle back to Good condition if it wasn't already.

△ **SOFT SIGNAL**

TicketController.php:403-407

07 Adding a sub-issue mid-ticket is limited to Custodian or Maintenance Personnel — never Admin, since it requires firsthand contact with the vehicle.

× **HARD BLOCK**

TicketController.php:427-450

08 Of those roles, only the ticket's own assigned Custodian, or a mechanic already on one of its sub-issues, may add a new one.

× **HARD BLOCK**

TicketController.php:446-450

09 Sub-issues can only be added while the ticket is Active — once Closed, a new discovery needs its own fresh ticket.

× **HARD BLOCK**

TicketController.php:452

10 Adding a sub-issue with a linked Issue Report flips that report to "In Maintenance."

△ **SOFT SIGNAL**

TicketController.php:474-476

01 Only Admin may assign a mechanic to a sub-issue.

✖ **HARD BLOCK**

TicketController.php:493

02 A mechanic can only be assigned while the ticket is Active and the sub-issue is still Open.

✖ **HARD BLOCK**

TicketController.php:496-497

03 The assignee must hold the Maintenance Personnel role and belong to the vehicle's barangay.

✖ **HARD BLOCK**

TicketController.php:508-509

04 Reassigning a mechanic is Admin-only, requires the ticket Active and the sub-issue "Under Repair."

✖ **HARD BLOCK**

TicketController.php:546,549-550

05 The new mechanic must qualify the same way as an initial assignment, and can't be the mechanic already on it.

✖ **HARD BLOCK**

TicketController.php:558-560

06 Reassigning a mechanic requires a reason, with both the old and new mechanic recorded/notified.

Why — explicitly so a repair on a critical vehicle (e.g. the only ambulance) doesn't freeze if the mechanic goes out sick.

✖ **HARD BLOCK**

TicketController.php:538-542,552-555

07 Only Maintenance Personnel may log repairs, and only the specific mechanic assigned to that sub-issue.

✖ **HARD BLOCK**

TicketController.php:696,699

08 Repairs can only be logged while the ticket is Active and the sub-issue is "Under Repair."

✖ **HARD BLOCK**

TicketController.php:700-701

09 Logging repairs auto-routes verification to the ticket's current custodian and moves the sub-issue to "For Inspection."

⚠ **SOFT SIGNAL**

TicketController.php:717-718

10 Supplying an estimated return date while logging repairs updates the vehicle's ETA.

⚠ **SOFT SIGNAL**

TicketController.php:729-731

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- 01 A Closed or Cancelled ticket's custodian can never be reassigned — finished work is left alone so the audit trail stays clean.
- × **HARD BLOCK** TicketController.php:623-627
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- 02 The new custodian must hold the Custodian role, belong to the vehicle's barangay, and can't be the one already on the ticket.
- × **HARD BLOCK** TicketController.php:637-639
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- 03 Reassigning the custodian requires a reason, for the audit trail.
- × **HARD BLOCK** TicketController.php:629-634
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- 04 Reassignment cascades to any sub-issue currently "For Inspection" — its verification moves to the new custodian too, so nothing is stranded on the old one.
- △ **SOFT SIGNAL** TicketController.php:648-656
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- 05 This action can only change who the custodian is — never status, verdicts, costs, or sub-issue content — a documented scope limit to keep it from rewriting history.
- × **HARD BLOCK** TicketController.php:611-615

- 01 Only Custodian role may verify a repair, and only the specific custodian the verification was assigned to.
× HARD BLOCK TicketController.php:755,758
- 02 Verification can only happen while the ticket is Active and the sub-issue is "For Inspection."
× HARD BLOCK TicketController.php:759-760
- 03 Verification requires an actual functional-test checklist with pass/fail per item — gated to Custodian specifically so the tester is independent of whoever did the repair.
Why — "don't grade your own homework."
× HARD BLOCK TicketController.php:762-775
- 04 Approving requires the custodian to explicitly attest they actually operated/tested the vehicle.
× HARD BLOCK TicketController.php:779-784
- 05 Cannot Approve if any checklist item failed — must Reject instead so the mechanic can redo the work.
× HARD BLOCK TicketController.php:785-791
- 06 Approving moves the sub-issue to "For Confirmation"; rejecting bounces it back to "Under Repair."
△ SOFT SIGNAL TicketController.php:793-803
- 07 Confirming/reworking a sub-issue is Admin-only, and only works while the ticket is Active and the sub-issue is "For Confirmation."
× HARD BLOCK TicketController.php:837,840-841
- 08 Confirming resolves the linked Issue Report (if any) and writes a permanent maintenance-record ledger entry — the single complete history of every confirmed fix.
△ SOFT SIGNAL TicketController.php:861-888
- 09 Once every sub-issue reaches Done, Admins get notified the ticket is ready to close — a notice, not a block.
△ SOFT SIGNAL TicketController.php:901-909
- 10 Rejecting at confirmation sends the sub-issue back to Under Repair and wipes its verification verdict, distinct from reopening a Closed ticket.
△ SOFT SIGNAL TicketController.php:910-924

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Un-confirming a Done sub-issue is Admin-only, only works on a Done sub-issue whose verdict was "Confirmed," and only while the ticket is Active — it restarts verification from "For Inspection."

× **HARD BLOCK**

TicketController.php:943-967

- 01 Only Admin may close a ticket, and only an Active one — an Open ticket stuck pre-inspection can't be closed (custodian reassignment exists to unstick that).
× HARD BLOCK TicketController.php:993,995
- 02 "Decision-close" with any sub-issue still unresolved requires both a mandatory reason and an explicit fit-for-service yes/no.
× HARD BLOCK TicketController.php:1012-1022
- 03 Every unresolved sub-issue at decision-close time is swept into "Deferred," each carrying the shared reason and its own breadcrumb Issue Report.
△ SOFT SIGNAL TicketController.php:1033-1041
- 04 A clean close (everything already Done/Deferred) always returns the vehicle to service; a decision-close honors whatever the Admin explicitly said.
△ SOFT SIGNAL TicketController.php:1025-1028
- 05 The originating Issue Report only resolves on a fit-for-service close, and never if that exact report is the one just deferred.
△ SOFT SIGNAL TicketController.php:1052-1056
- 06 If not fit for service, closing the ticket does NOT free the vehicle — it stays Under Maintenance even though the ticket itself is Closed.
△ SOFT SIGNAL TicketController.php:1061-1072
- 07 Vehicle status can only flip back to Available through one choke point that checks no OTHER ticket on it is still open — one closed ticket can't free a vehicle another ticket still has claimed.
× HARD BLOCK TicketController.php:1314-1336
- 08 Closing archives the ticket permanently — this is what locks it from later deletion, custodian reassignment, or new sub-issues.
△ SOFT SIGNAL TicketController.php:1059
- 09 Deferring a single sub-issue (outside a full close) is Admin-only, requires the ticket Active, the sub-issue unresolved, and a reason.
× HARD BLOCK TicketController.php:1105-1117
- 10 Cannot defer a sub-issue that's already "For Confirmation" — the Custodian already verified it works, so Confirm or Reopen are the only choices left that make sense.
× HARD BLOCK TicketController.php:1110-1113

- 11 Deferring always creates or resurfaces a breadcrumb Issue Report, so the unfixed defect is never simply forgotten.

△ **SOFT SIGNAL**

TicketController.php:1358-1387

- 01 Only Admin may cancel a ticket, and it can't already be Closed or Cancelled.

✖ HARD BLOCK TicketController.php:1143,1145
- 02 Cancelling resets every linked Issue Report — ticket-level and every sub-issue's — back to Pending.

⚠ SOFT SIGNAL TicketController.php:1157
- 03 Only Admin may restore a cancelled ticket, and only a Cancelled one qualifies.

✖ HARD BLOCK TicketController.php:1173,1175
- 04 Restoring lands back on Active if it was ever inspected, otherwise Open — and resets linked issues to "In Maintenance."

⚠ SOFT SIGNAL TicketController.php:1177,1185
- 05 A Closed ticket can never be cancelled, so uncanceled can never accidentally target genuinely completed work.

✖ HARD BLOCK TicketController.php:1166-1170
- 06 Only Admin may delete a ticket, and a Closed ticket can NEVER be deleted — it is permanent, full stop.

✖ HARD BLOCK TicketController.php:1208,1210
- 07 Deleting resets linked Issue Reports back to Pending.

⚠ SOFT SIGNAL TicketController.php:1217
- 08 If the deleted ticket had at least one genuinely completed (Done) sub-issue, it's archived as recoverable first — "the accidental click on unfinished work" case; a ticket with zero completed work is just discarded.

⚠ SOFT SIGNAL TicketController.php:1194-1221
- 09 Deleting a ticket cascades to delete its sub-issues at the database level.

⚠ SOFT SIGNAL TicketController.php:1223
- 10 Only Admin may reopen an archived (deleted) ticket, and only one tagged "Deleted" — a "Closed" archive is permanently locked and refused.

✖ HARD BLOCK TicketController.php:1241,1243
- 11 Reopening is blocked if it would recreate a duplicate main issue against an already-open ticket on the same vehicle.

✖ HARD BLOCK TicketController.php:1247-1253

- 12 Reopening recreates the ticket as a brand-new record from the archived snapshot and re-flips any linked issues back to "In Maintenance."

△ **SOFT SIGNAL**

TicketController.php:1255-1305

VISIBILITY & ACCESS — 3

- 01 The duplicate-check helper (open tickets on a vehicle) is Admin-only and purely informational — a "Layer 2" soft warning shown while picking a vehicle on Create Ticket, distinct from the hard duplicate block at creation.

△ **SOFT SIGNAL**

TicketController.php:117-133

- 02 Browsing the ticket archive is Admin-only.

× **HARD BLOCK**

TicketController.php:137

- 03 Ticket list visibility is role-scoped: a Custodian sees only tickets they're assigned to; Maintenance Personnel sees only tickets with a sub-issue assigned to them; a dual-role account sees the union; Admin sees everything.

△ **SOFT SIGNAL**

TicketController.php:63-72

Part 3 Auth, Users & Administration

45 rules · AuthController · UserController · SuperAdminController · ConcernReportController · Middleware

Registration and the first-admin bootstrap, login/session handling, self-service profile, Admin's user management, impersonation, Super Admin's account administration, public concern reports, and the two access-control middlewares underneath all of it.

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- 01 A self-registered account is inactive by default and needs Admin approval before it can log in.
- × **HARD BLOCK** AuthController.php:126-128
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- 02 The first person to register under a given barangay automatically becomes its Admin — active immediately, no approval, no staff code needed.
- Why — each barangay is its own tenant and needs a bootstrap admin since nobody exists yet to approve one.*
- △ **SOFT SIGNAL** AuthController.php:93,105-113,126-127
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- 03 Anyone registering under a barangay that already has a user must supply that barangay's correct Staff Registration Code, or registration is rejected outright.
- Why — the actual gate keeping a stranger from ever reaching the Admin's approval queue.*
- × **HARD BLOCK** AuthController.php:98-99,106-111
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- 04 A non-first registrant must pick a requested role of Custodian or Maintenance Personnel — there is no public path to registering as Admin.
- × **HARD BLOCK** AuthController.php:98
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- 05 Registration for a brand-new barangay locks the Barangay row for the duration of the "am I first" check plus user creation, so two simultaneous registrants can't both become Admin.
- × **HARD BLOCK** AuthController.php:88-89
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- 06 Exactly one of barangay (by dropdown) or barangay name (free text) must be supplied, and a dropdown pick must belong to the selected city.
- × **HARD BLOCK** AuthController.php:53-63
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- 07 The registration-status preview endpoint is read-only and never creates a barangay or user.
- △ **SOFT SIGNAL** AuthController.php:152-171

LOGIN & SESSION — 5

- 01 Wrong email or password returns 401 — no account created, nothing revealed about which part was wrong.
× HARD BLOCK AuthController.php:188-192
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- 02 A deactivated account can't log in even with correct credentials.
× HARD BLOCK AuthController.php:194-198
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- 03 Every issued token is tagged with all of the user's current roles, so its abilities match actual account access.
△ SOFT SIGNAL AuthController.php:200-202
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- 04 A deactivated account's existing token stops working on the very next request, not just on future logins.
Why — previously `is_active` was only checked at login, so a deactivated user's live session kept working indefinitely.
× HARD BLOCK EnsureUserIsActive.php:21-23
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- 05 Logout only revokes the current request's token, not every session the account has open.
△ SOFT SIGNAL AuthController.php:333

SELF-SERVICE PROFILE — 3

- 01 Changing your own password requires correctly supplying the current password first.
× HARD BLOCK AuthController.php:471-475
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- 02 Self-service profile editing covers name/email/phone/address/photo — but never role or password, which stay on their own Admin-reviewed and old-password-verified paths respectively.
× HARD BLOCK AuthController.php:423-440
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- 03 Email must stay unique across accounts when self-updating a profile.
× HARD BLOCK AuthController.php:436

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- 01 Only Admin can access any user-management endpoint at all.
- ✗ **HARD BLOCK** UserController.php:18-21
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- 02 An Admin can only manage users within their own barangay — reaching for a user in another barangay returns 404, not 403, so it reads as "no such user" rather than confirming someone else's account exists there.
- ✗ **HARD BLOCK** UserController.php:24-33
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- 03 A new user created directly by an Admin always lands in the creating Admin's own barangay, regardless of what the client sends.
- ✗ **HARD BLOCK** UserController.php:96-98
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- 04 A user created directly by an Admin is active immediately — no approval gate, since the Admin creating them IS the approver.
- △ **SOFT SIGNAL** UserController.php:95
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- 05 An Admin can never deactivate their own account.
- ✗ **HARD BLOCK** UserController.php:149
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- 06 Deactivating a user immediately revokes every one of their existing tokens, not just future logins.
- △ **SOFT SIGNAL** UserController.php:151-154
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- 07 approved_at is stamped only on the very first activation — re-activating an already-approved user never overwrites it.
- △ **SOFT SIGNAL** UserController.php:173-178
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- 08 The optional role override on activation is only meaningful for a first-time approval (confirming/correcting the self-requested role) — not expected on a plain re-activation.
- △ **SOFT SIGNAL** UserController.php:164-171
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- 09 Staff Registration Code endpoints are scoped to the requesting Admin's own barangay only.
- ✗ **HARD BLOCK** UserController.php:190-204
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- 10 The "pending" filter only matches never-yet-approved accounts — a previously-approved account that's since been deactivated does NOT count as pending.
- △ **SOFT SIGNAL** UserController.php:69-72

01 Impersonation is dev-only for an Admin — it 404s outright outside local/testing environments.

× **HARD BLOCK**

AuthController.php:226-232

02 Super Admin is exempt from that dev-only restriction — impersonation works for them in production, since real account support/recovery is exactly what the role is for.

△ **SOFT SIGNAL**

AuthController.php:228-230

03 Only an Admin, a Super Admin, or a token that already carries the "impersonated" ability may impersonate or list candidates.

Why — a genuine Custodian's own login token can never carry that ability — it's only ever granted by impersonating in the first place.

× **HARD BLOCK**

AuthController.php:244-249

04 The impersonation target must be an active account.

× **HARD BLOCK**

AuthController.php:300

05 Impersonation deliberately crosses barangay boundaries — for an Admin it's dev-only so tenant isolation doesn't apply; for Super Admin, cross-barangay reach is the whole point.

△ **SOFT SIGNAL**

AuthController.php:256-261,298-299

06 Every impersonation is written to the activity log, tagged "Super Admin" or "Dev Tools" depending on who's acting.

△ **SOFT SIGNAL**

AuthController.php:304-312

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- 01 Every Super Admin endpoint requires the Super Admin role specifically.
- × **HARD BLOCK** SuperAdminController.php:27-29
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- 02 A Super Admin can never deactivate their own account — the same self-protection rule as a barangay Admin.
- × **HARD BLOCK** SuperAdminController.php:109
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- 03 Deactivating a user as Super Admin also immediately revokes all of that user's tokens.
- △ **SOFT SIGNAL** SuperAdminController.php:112
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- 04 approved_at is still a one-time stamp when Super Admin activates a user, same as the barangay-Admin path.
- △ **SOFT SIGNAL** SuperAdminController.php:100
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- 05 A Super Admin can change any user's role, in any barangay — the documented mechanism for recovering an "orphaned" barangay by promoting an existing user there to Admin.
- △ **SOFT SIGNAL** SuperAdminController.php:118-134
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- 06 The barangay and user listings visible to Super Admin deliberately span every barangay — the one place in the system cross-tenant visibility is intentional, not a leak.
- △ **SOFT SIGNAL** SuperAdminController.php:43-47,66-69,153-157
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- 07 Every Super Admin account action is written to the activity log under module "Super Admin."
- △ **SOFT SIGNAL** SuperAdminController.php:32-41
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- 08 A Super Admin can only ever reach account-administration endpoints — never a barangay's fleet data — enforced by having no barangay_id to scope by, backstopped by middleware below.
- × **HARD BLOCK** SuperAdminController.php:13-22

CONCERN REPORTS — 4

- 01 Submitting a concern report requires no login at all — fully public, since the reporter may be an ordinary resident.
- △ **SOFT SIGNAL** ConcernReportController.php:14-18
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- 02 Concern type must be one of three fixed values: Barangay Inactive, Suspected Fake Staff, or Other.
- × **HARD BLOCK** ConcernReportController.php:11,21
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- 03 Description is required; barangay name and reporter identity are both optional — a report can be filed anonymously without pinpointing an exact barangay.
- × **HARD BLOCK** ConcernReportController.php:20-26
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- 04 There is no self-service way to edit or withdraw a submitted report — only Super Admin can resolve or reopen one.
- △ **SOFT SIGNAL** ConcernReportController.php (store only)

MIDDLEWARE — PLATFORM-WIDE GATES — 2

- 01 A Super Admin's requests are allowlisted to only superadmin/*, impersonate/*, logout, user, profile, and profile/password — everything else is refused with "Super Admin accounts are limited to account administration and cannot access fleet data."
- Why — ordinary fleet endpoints would otherwise silently leak every barangay's data to an account with no barangay_id to scope against — a single allowlist instead of a role check bolted onto dozens of controllers.*
- × **HARD BLOCK** RestrictSuperAdminScope.php:22-23,38-42
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- 02 On every request, if the authenticated user's account has since gone inactive, the request is refused with 401 — closing the exact gap where a deactivated user's still-valid token kept working until it happened to expire.
- × **HARD BLOCK** EnsureUserIsActive.php:17-23